

August 2023

INTI International College Penang

School of Computing

5003CEM

Date: 04 December 2023 (Monday)

Advanced Algorithms

Time: 12.00pm – 02.00pm

Instructions to candidates

Time allowed: **2 hours 0 minutes**

This is a **Closed Book** Examination

Answer: All Questions

The total number of questions in this paper: 4

All questions carry equal marks

Start each question **on a new page** and carefully identify your answers with the correct question number

For this examination you will be supplied with the following:

1 Answer Book/s

You must hand this question paper in at the end of the examination.

Answer all questions

Each question is worth 25 marks

1. a) Given the following list of numbers:

23, 57, 48, 17, 33, 39, 42, 28, 9, 11

Construct a AVL tree. Show all the steps and if rotation is required show the tree before and after the rotation.

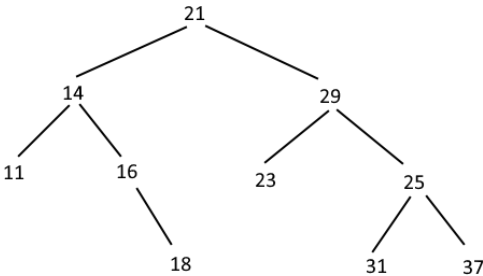
(10 marks)

b) Given the BST and Node classes defined as follow:

C++:	Python:
<pre>struct Node{ public: int value; Node* left; Node* right; }; class BST{ private: Node* root; }</pre>	<pre>class Node(object): def __init__(self,value): self.value = value self.left = None self.right = None class BST(object): def __init__(self): self.root = None</pre>

Write a function for the BST class named `find_next()` that will take an integer value `num` and return the smallest value in the binary tree that is strictly larger than `num`. If no such value exists, return `num` itself.

For example in the binary search tree below, `find_next(29)` will return 31.



Note: you can provide your answer in either C++ or Python.

(10 marks)

- c) Compare between the efficiency of searching for a value in a binary search tree versus in a AVL tree. Highlight which is more efficient and explain why.

(5 marks)

2. Refer to the following list of number for this question:

18, 34, 56, 40, 11, 59, 13, 43

- a) (i) Construct a min heap using the list of numbers given above.

(4 marks)

- (ii) Give the result of an in-order traversal of the heap produced in part (a)(i).

(4 marks)

- (iii) Demonstrate the process of sorting the list of number given above, in ascending order, using the min heap from part (a)(i).
Show all the working steps.

(8 marks)

- b) Consider the following partially filled hash table and hash function:

$$h(k) = (k + 3)^2 \bmod 13$$

[0]	23
[1]	
[2]	
[3]	14
[4]	
[5]	
[6]	
[7]	
[8]	
[9]	
[10]	
[11]	
[12]	

- (i) Insert the numbers from Question 2 earlier into the hash table above using the hash function given. In the event of a collision, use linear probing to handle it.

Show all the working steps and indicate clearly when a collision happens and illustrate how it is resolved.

(6 marks)

3. a) (i) A file contains data which is made up of only 6 letters, encoded using a fixed length 3-bit code word.

Find the optimal code using Huffman code given that the 6 letters have the following frequency:

Letter	Frequency
A	11,000
B	17,000
C	23,000
D	10,000
E	21,000
F	12,000

- (ii) Calculate the total saving in bits that can be achieved by using the optimal code found in part (a)(i). (9 marks)

(1 mark)

- b) The travelling time taken from one location to another is shown in the following adjacency list:

Location	Adjacency List
P	R(7), S(17)
Q	S(12), T(11)
R	Q(13), T(6)
S	P(8), R(10)
T	S(3)

- (i) Illustrate the adjacency list in a diagram form. (5 marks)

- (ii) Use Dijkstra's algorithm to find the shortest path to all the other locations starting from P. Your final answer should include the shortest path as well as the length of the path.

(10 marks)

- (ii) Name and explain **ONE (1)** method of resolving collisions in a hash table. (other than linear probing).

(3 marks)

4. a) Express the order of growth of the following expressions using big O-notation. (4 marks)
- (i) $2n^3 + 23n^4 + 15n + 1$
 - (ii) $n^{1024} + 2^n$
 - (iii) $30n + 30n \log n$
 - (iv) $12^n + n!$
- b) For each of the scenarios given below, identify the worst case big-O running time for each. Provide an explanation to support your answers as well.
- (i) Copying all the numbers from a linked list and pushing them into a stack. (3 marks)
 - (ii) Finding the maximum value in a binary search tree. (3 marks)
 - (iii) Moving all the elements from a binary min heap into a array, arranging them in ascending order. (3 marks)
- c) Based on the initial unsorted list:
- 15, 36, 24, 52, 1, 12, 59, 47
- Show the steps to sort the numbers using Merge Sort. (6 marks)
- d) In the context of concurrent programming, explain what is a deadlock and describe some ways in which it can be avoided. (6 marks)

~THE END~